

## RS003 Week 3 — English Worksheet

**Topic:** Build the Ship Body · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week you think about the code that draws the first piece of your spaceship: a **body** that sits in the middle-bottom of the screen. You read code that draws shapes, stores important numbers as **constants**, and finds the centre of the screen with **integer division** ( `//` ). You will read it, fix it, and explain it.

Words to know: **constant**, **rectangle**, **circle**, **integer division** ( `//` ), **centre**

### 1 · Predict 🧠

Read each piece of code. Just from reading it, write what you think it does. Do not run it.

```
WIDTH = 800
CENTER_X = WIDTH // 2
```

What number is `CENTER_X` ? Why use `//` instead of `/` here?

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```
pygame.draw.rect(screen, CYAN, (CENTER_X - 20, HEIGHT - 80, 40, 50))
```

The four numbers are (left, top, width, height). Where on the screen does this rectangle appear, and how big is it?

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```
pygame.draw.circle(screen, WHITE, (CENTER_X, HEIGHT - 90), 8)
```

**What shape, where, and how big? What does the last number 8 control?**

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```
screen.fill(SPACE)
pygame.draw.rect(screen, CYAN, (CENTER_X - 20, HEIGHT - 80, 40, 50))
pygame.display.flip()
```

**In what order do these three lines run? What would you see if `screen.fill(SPACE)` came AFTER the rectangle instead of before it?**

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## 2 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it on paper, then explain why the original was wrong.

**Bug A** — The ship body should sit in the **centre** across the screen. Right now it is glued to the left edge.

```
pygame.draw.rect(screen, CYAN, (0, HEIGHT - 80, 40, 50))
```

**Hint:** use `CENTER_X` to place it in the middle.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — `CENTER_X` is supposed to be the middle of the screen, but it comes out as a number with a decimal point and the drawing looks off.

```
CENTER_X = WIDTH / 2
```

**Hint:** screen positions must be whole numbers.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug C** — The body is supposed to sit near the **bottom**, where the player ship belongs. Right now it is drawn near the top of the screen.

```
pygame.draw.rect(screen, CYAN, (CENTER_X - 20, 80, 40, 50))
```

**Hint:** in Pygame, Y = 0 is the **top**. The bottom is near **HEIGHT**.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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### 3 · Explain the Code

Read this working example line by line. Then answer the questions below by reading only — you do not need to run it.

```
import pygame
pygame.init()

WIDTH = 800
HEIGHT = 600
CENTER_X = WIDTH // 2

screen = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("Space Shooter")

SPACE = (10, 12, 40)
CYAN = (80, 220, 255)

clock = pygame.time.Clock()
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    screen.fill(SPACE)
    pygame.draw.rect(screen, CYAN, (CENTER_X - 20, HEIGHT - 80, 40, 50))
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

**WIDTH**, **HEIGHT**, **SPACE**, and **CYAN** are all constants. What is each one used for?

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The rectangle's left value is `CENTER_X - 20` , not `CENTER_X` . Why subtract 20? What would happen if you used `CENTER_X` on its own?

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What does the `while running:` loop do? What makes the loop stop?

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Why is the cyan rectangle drawn in the middle across the screen? Point to the exact lines that make this happen.

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Is the rectangle near the top or near the bottom of the window? How can you tell from the numbers?

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## 4 · Explain Your Lesson Code 🎥

In today's live lesson you wrote code with your teacher to draw your ship body. Now explain that code in a short phone video. You may show it running on the screen. Try to use these words: **constant, rectangle, circle, integer division, centre.**

1. Show your ship body sitting in the middle-bottom of the screen.
2. Read your `CENTER_X` line out loud and say what `//` does.
3. Point at your rectangle's four numbers and name each one (left, top, width, height).
4. Say why `Y = 0` is the top, not the bottom.

**Write what you will say in your video. Plan it here before you record.**

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**Submit** ☒

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.